Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 1: Approved Solutions

Statements, Truth Values, and Open Sentences

Covers Chartrand & Zhang, Chapter 0 and Section 1.1

Status of this set

This set is **ungraded practice**. It is not submitted for a grade, and the approved solutions are released with it so you can check your own work. Your first *graded* problem set covers Sections 1.1–1.2 and will be posted on Canvas.
Work the problems on your own paper — there is deliberately no space to write here, so that you can attempt a problem more than once. Practice only works if you attempt every problem **before** reading any solution; see "The one rule that matters" below.

Part I: How to Study for This Course

This course grades your reasoning, not your answer

In CS 204 a program either produced the right output or it did not. Here, a correct final answer with no reasoning earns very little credit, and a wrong final answer with sound, clearly written reasoning earns a great deal. The written argument *is* the work.

Get used to this now, in Week 1, while the content is easy. The habits are the hard part, not the mathematics.

The one rule that matters

Attempt first, then check

Work every problem completely before you look at the solution — including the ones you are stuck on. Write down where you got stuck and *why*.
Reading a solution you have not attempted feels productive and is not. You will recognize every step, conclude that you understood it, and then be unable to reproduce any of it on a quiz. Recognition is not the same as recall, and only one of the two is graded.

A workable weekly cycle

1. **Before class:** read the assigned pages with a pen, working each of the book's examples yourself before reading its solution. Expect roughly **ten minutes per page**, not one — if you are reading a mathematics text at the speed you read an article, you are not reading it.

2. **During class:** fill in the guided notes. The parts you write are the parts you retain.
3. **Within 24 hours:** do the practice problems. Memory decays fastest on day one.
4. **Then, and only then:** check against the approved solutions. Mark each problem correct, partially correct, or wrong — and for anything not fully correct, write one sentence naming what you missed.
5. **Office hours:** bring the specific problems you marked, with your attempt on paper. "I don't get it" is hard to help with; "I got stuck at this line, because I wasn't sure whether this counts as a statement" takes about ninety seconds to resolve.

What a complete written answer looks like

Consider: *Is "$x + 2 = 7$" a statement? Explain.*

No credit	"No."
Partial	"No, it's an open sentence."
Full	"No. It contains the variable x, and its truth value depends on the value of x: if $x = 5$ it is true, and if $x = 3$ it is false. Because it does not have one definite truth value on its own, it is an open sentence rather than a statement. It becomes a statement once x is assigned a value."

The full answer names the relevant feature (the variable), shows the consequence with concrete cases, applies the definition, and states the conclusion. Every graded answer in this course should do those four things.

Collaboration and generative AI, in practice

You may discuss *approaches* with classmates. You must write your final solution yourself, in your own words, from your own understanding.

For graded work, generative AI is limited to grammar, formatting, and LaTeX help — not generating, structuring, or substantially revising your answers. The reason is specific to this course: the skill being assessed is *producing the reasoning*, so outsourcing it removes the only thing the grade measures. You may be asked to explain any submitted step, and being unable to do so is treated as not having done the work.

Check this before you hand in any written work

- Did I answer the question that was actually asked?
- Is every claim justified, or did I call something obvious to avoid explaining it?
- Could a classmate who missed class follow my reasoning without asking me anything?
- Did I apply the *definition*, or did I appeal to intuition?
- Are these complete sentences? Mathematics is written in prose, not in fragments.
- Did I state my conclusion explicitly, rather than leaving it to be inferred?

Five mistakes almost everyone makes in Week 1

1. **"Statement" means "something I can check."** It does not. A sentence is a statement because it *has* a definite truth value, not because you know that value.
2. **Confusing "unknown" with "neither true nor false."** An unsolved conjecture is still a statement.
3. **Calling an open sentence false.** "$x + 2 = 7$" is not false; it has no truth value at all until x is fixed.
4. **Giving the answer with no reasoning.** See the table above.
5. **Reading "or" as exclusive.** In mathematics "P or Q" is true when both hold.

Part II: Practice Problems

Section A — Is it a statement?

Classify each sentence as exactly one of the following, and give a one-sentence reason.

true statement	definitely true
false statement	definitely false
statement, truth value unknown	definitely one or the other, but nobody currently knows which
open sentence	contains a variable; no truth value until the variable is fixed
not a statement	not the kind of sentence that can be true or false at all

1. 17 is a prime number.

Approved solution

True statement. It is declarative, and 17 has no positive divisors other than 1 and itself.

2. Is 51 a prime number?

Approved solution

Not a statement. It is a question, so it is not declarative and cannot be assigned a truth value.

3. 51 is a prime number.

Approved solution

False statement. It is declarative and false, since $51 = 3 \cdot 17$.
Note the contrast with #2. The mathematical content is identical; only the grammatical form changed. Whether a sentence is a statement is a question about the *sentence*, not about the subject matter.

4. n is a prime number.

Approved solution

Open sentence. It contains the variable n and has no truth value until n is assigned a value. If $n = 17$ it becomes true; if $n = 51$ it becomes false.

5. Please submit your homework through Canvas.

Approved solution

Not a statement. It is a request or command, not a declarative sentence.

6. $\sqrt{2}$ is a rational number.

Approved solution

False statement. It is declarative and false; $\sqrt{2}$ is irrational. (We will prove this in Chapter 3 using proof by contradiction.)

7. There is intelligent life on a planet outside our solar system.

Approved solution

Statement, truth value unknown. Either such life exists or it does not, so the sentence has a definite truth value. Nobody currently knows which. Our ignorance does not change its status as a statement.

8. This sentence is false.

Approved solution

Not a statement. Suppose it were true. Then what it asserts holds, so it is false — a contradiction. Suppose instead it were false. Then what it asserts fails, so it is true — also a contradiction. No consistent truth value can be assigned, so it fails the definition, which requires the sentence to be true or false.

9. $x^2 \geq 0$, where x is a real number.

Approved solution

Open sentence. It contains the free variable x.
This one is worth pausing on. It happens to be true for *every* real x, which tempts people to call it a true statement. But as written it still has a free variable, so it is an open sentence. It becomes a true statement only once the variable is quantified: "for every real number x, $x^2 \geq 0$" *is* a true statement. Quantifiers are Section 3.1.

10. Discrete mathematics is more useful than calculus.

Approved solution

Not a statement. "More useful" is not defined here — useful for what, measured how? Without an agreed criterion the sentence has no objective truth value.
Contrast this with #7. Both are sentences we cannot currently settle, but for different reasons: #7 has a definite answer we do not know, while this one has no definite answer to know.

11. The 10^{100}-th decimal digit of π is 7.

Approved solution

Statement, truth value unknown. The digit is some specific one of 0 through 9, so the sentence is definitely true or definitely false. Computing it is far beyond current reach — but computability is irrelevant to the definition.

12. Every even integer greater than 2 is the sum of two prime numbers.

Approved solution

Statement, truth value unknown. This is Goldbach's conjecture, open since 1742. It has been verified for every even number tested by computer, which is evidence but not proof. It is either true or false; nobody has established which.

Section B — Open sentences

13. Let $P(x) : x^2 - 5x + 6 = 0$, where x is an integer.

(a) Is $P(2)$ true or false?

(b) Is $P(4)$ true or false?

(c) Find *all* integers x for which $P(x)$ is true.

Approved solution

(a) True. $2^2 - 5(2) + 6 = 4 - 10 + 6 = 0$.
(b) False. $4^2 - 5(4) + 6 = 16 - 20 + 6 = 2 \neq 0$.
(c) $x = 2$ and $x = 3$. Factoring, $x^2 - 5x + 6 = (x - 2)(x - 3)$, and a product of two integers is zero exactly when one of the factors is zero. So the expression equals zero precisely when $x = 2$ or $x = 3$.

14. Let $Q(n) : n^2 + n$ is even, where n is an integer.

(a) Evaluate $Q(3)$, $Q(4)$, and $Q(-5)$.

(b) Based on your results, what do you suspect about $Q(n)$ in general?

(c) Give a reason your suspicion is correct. (You do not need a formal proof yet.)

Approved solution

(a) $Q(3)$: $9 + 3 = 12$, even, so **true**. $Q(4)$: $16 + 4 = 20$, even, so **true**. $Q(-5)$: $25 + (-5) = 20$, even, so **true**.
(b) $Q(n)$ appears to be true for every integer n.
(c) Factor: $n^2 + n = n(n + 1)$. These are two consecutive integers, so exactly one of them is even. A product with an even factor is even, so $n(n + 1)$ is always even.
Why part (c) is the real question. Parts (a) and (b) only show the claim holds in three cases. Three cases are not a proof — see the next problem for exactly why.

15. Let $R(x) : 2^x > x^2$, where x is a positive integer.

(a) Evaluate $R(x)$ for $x = 1, 2, 3, 4, 5, 6$.

(b) A classmate checks $R(1)$, sees that it is true, and concludes that $R(x)$ is true for every positive integer. What is wrong with that reasoning?

(c) For which positive integers does $R(x)$ appear to be true?

Approved solution

(a)

x	2^x	x^2	$R(x)$
1	2	1	true
2	4	4	false
3	8	9	false
4	16	16	false
5	32	25	true
6	64	36	true

(b) Checking one case — or any finite number of cases — shows only that the open sentence is true *for those values*. It says nothing about the others. Here the classmate would be badly wrong: the very next case, $x = 2$, is false.
(c) True for $x = 1$, false for $x = 2, 3, 4$, and true for every $x \geq 5$.
Why this problem is in the set. The pattern is not monotone at the start. Anyone who tested only $x = 1$, or only $x = 5$ and $x = 6$, would draw the wrong conclusion. This is the single best argument for why this course insists on proof rather than sampling.

16. Let $S(x, y) : x + y = 10$, where x and y are positive integers. How many pairs (x, y) make $S(x, y)$ true? List them.

Approved solution

Nine pairs: $(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)$.
Once x is chosen, y is forced to be $10 - x$. For y to be a positive integer we need $1 \leq x \leq 9$, giving nine choices for x and therefore nine pairs.
Note: (x, y) is ordered, so $(3, 7)$ and $(7, 3)$ count separately.

Section C — Counting distinct possibilities

17. A PIN is exactly four digits, each from 0 through 9, and digits may repeat. How many PINs are possible?

Approved solution

$10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = \mathbf{10{,}000}$. Each of the four positions is chosen independently from ten options.

18. Same as above, except no digit may be repeated. How many PINs are possible?

Approved solution

$10 \cdot 9 \cdot 8 \cdot 7 = \mathbf{5040}$. The first position has ten options; each later position has one fewer, since the digits already used are unavailable.

19. How many binary strings have length exactly 10?

Approved solution

$2^{10} = \mathbf{1024}$. Each of the ten positions is independently `0` or `1`.

20. A password is exactly 8 characters, each chosen from the 26 lowercase letters, 26 uppercase letters, and 10 digits.

(a) How many passwords are possible?

(b) Roughly how many times larger is this than the number of four-digit PINs?

Approved solution

(a) There are $26 + 26 + 10 = 62$ choices per character and 8 characters, giving $62^8 = \mathbf{218{,}340{,}105{,}584{,}896}$, or about 2.18×10^{14}.
(b) About $2.18 \times 10^{14} \div 10^4 \approx 2.18 \times 10^{10}$ times larger — roughly **twenty billion times**.
Why this matters. Both sets are finite and discrete. "Discrete" says nothing about size. The entire practical difference between a weak and a strong credential is a counting argument.

21. Classify each quantity as discrete or continuous, and give a brief reason for any two of them.

(a) The number of edges in a network.

(b) The voltage on a wire.

(c) The number of comparisons a sorting algorithm performs.

(d) The temperature of a CPU.

(e) The set of valid IPv4 addresses.

Approved solution

(a) Discrete — edges are counted in whole units; there is no such thing as 3.5 edges.
(b) Continuous — voltage varies smoothly and takes real values in a range.
(c) Discrete — comparisons are counted; the algorithm performs a whole number.
(d) Continuous — temperature varies smoothly over time.
(e) Discrete — there are exactly 2^{32} IPv4 addresses, a finite, separate, countable collection.
Accept any reasonable justification. Part (e) is the one to discuss: 2^{32} is a large number, and largeness is not continuity.

Section D — Precision in ordinary language and in code

22. A course policy reads: *An assignment receives credit if it is submitted before the deadline and passes the automated tests.* For each case, does the assignment receive credit?

(a) Submitted on time; passes the tests.

(b) Submitted on time; fails the tests.

(c) Submitted late; passes the tests.

(d) Submitted late; fails the tests.

Approved solution

(a) Yes. (b) No. (c) No. (d) No.
The word *and* requires both conditions to hold, so credit is given in exactly one of the four cases. Had the policy said *or*, credit would be given in cases (a), (b), and (c) — three of four. A single word changes the outcome in half the cases.

23. A security requirement reads: *A password is accepted only if it is longer than eight characters and contains at least one symbol.* A student writes:

```
if len(password) > 8 or has_symbol(password):
    accept()
```

(a) Give a specific password this code wrongly accepts.

(b) Correct the code.

Approved solution

(a) `"abcdefghij"` — ten characters, so the first condition holds and the `or` short-circuits to acceptance, even though the password contains no symbol. Any long symbol-free password works, as does any short password containing a symbol, such as `"a!"`.
(b)

```
if len(password) > 8 and has_symbol(password):
    accept()
```

The point. The requirement and the buggy code differ by one word. Reading the specification precisely is the whole job.

24. For each sentence, say whether "or" is being used inclusively (both may hold) or exclusively (exactly one holds), and explain how you can tell.

(a) "Bring a pen or a pencil to the exam."

(b) "Your entrée comes with soup or salad."

(c) "To enroll, a student must have completed CS 204 or received instructor permission."

Approved solution

(a) Inclusive. Bringing both violates nothing; the sentence sets a minimum, not a limit.
(b) Exclusive in ordinary use — the restaurant intends one, not both. Note that nothing in the grammar says so; we infer it from context and cost.
(c) Inclusive. A student with both CS 204 and instructor permission is certainly eligible.
The lesson. English "or" is settled by context, which is exactly the ambiguity mathematics cannot

tolerate. This is why $\vee$ is *defined* as inclusive and a separate symbol $\oplus$ exists for the exclusive case. Wednesday's class makes both precise.

Section E — Writing

25. In four to six sentences, explain to a classmate why "There are 25 students in this room" is a statement even before anyone counts, while "$x + 2 = 7$" is not a statement at all. Your answer should make clear that these two sentences fail to have a known truth value for *different reasons*.

Approved solution

Model answer. "There are 25 students in this room" is a declarative sentence with a definite truth value: the room contains some specific number of students, so the sentence is either true or false right now. What we lack is knowledge of which, and knowledge is not part of the definition of a statement — counting would reveal the truth value, not create it. By contrast, "$x + 2 = 7$" has no truth value at all, because x has not been given a value; it is true when $x = 5$ and false when $x = 3$. So the first sentence is a statement whose truth value we happen not to know, while the second is an open sentence with nothing to know yet. Assigning x a value turns the second into a statement; counting does not turn the first into one, because it already was one.
What earns full credit: the answer must locate the difference in *the sentences*, not in our knowledge. Any answer implying "both are unknown, so both are the same" misses the distinction the problem is testing.

Section F — Challenge (optional)

26. Consider the sentence "This sentence has five words."

(a) Is it a statement? If so, what is its truth value?

(b) Compare it with problem #8. Both sentences refer to themselves. Why is only one of them a problem?

Approved solution

(a) It is a **true statement**. Counting: *This*, *sentence*, *has*, *five*, *words* — five words.
(b) Self-reference alone is harmless. This sentence refers to a property of itself (its word count) that can be checked independently of its truth value, so no circularity arises. The liar sentence in #8 refers to *its own truth value*, and that is what creates the loop: computing the truth value requires already having it. The trouble is self-reference *about truth*, not self-reference as such.
Extension worth mentioning aloud: "This sentence has six words" is also a statement — a false one. Self-reference does not even prevent a definite truth value.

27. Suppose nobody in our classroom is over 200 years old. Consider: "Every student in this room who is over 200 years old has a Ph.D." Is this true, false, or neither? Defend your answer. (This one is genuinely contested on first encounter — we settle it formally in Section 1.3.)

Approved solution

True — and this is called *vacuous truth*.
The sentence claims that every member of a certain group has a property. That group is empty, so there is no student who could serve as a counterexample. A claim about all members of an empty

collection cannot be violated, and by convention (and by the truth table for implication, which we build in Section 1.3) such a claim is counted as true.
Most students want to answer "neither" or "meaningless" here, which is a reasonable instinct and the wrong answer. Note the payoff: this convention is what makes "every element of the empty list satisfies the condition" come out true in code, which is exactly what `all([])` returns in Python.

Self-check rubric

Score each written problem out of 4 and record the total.

Score	Level	Description
4	Complete	Correct conclusion, correct reason, stated in full sentences. Names the relevant feature and applies the definition to it.
3	Sound but thin	Correct conclusion and essentially correct reason, but the reasoning is compressed or a step is left implicit.
2	Right answer, weak support	Correct conclusion with a reason that is vague, circular, or appeals to intuition rather than the definition.
1	Answer only	Correct or near-correct conclusion with no reasoning.
0	Incorrect	Wrong conclusion, or reasoning that contradicts the definition.

Grading notes for the instructor

- **#9 and #22** are the two most valuable items in the set. Both target the misconception that a statement is "something checkable." If students miss these, revisit before quantifiers in Section 3.1.
- **#15** is the argument for the entire course. If nothing else lands, land this one: sampling cases is not proof, and here the pattern actively misleads.
- **#10 vs. #7** is the sharpest pair. Both are unsettled; only one has a truth value to settle. Expect this distinction to need a second pass.
- **#24** may be worth ten minutes at the board rather than assigned reading — students argue about it productively.
- **#12** previews the counting content of Chapter 8. Do not let it become a calculator exercise; the reasoning is the multiplication principle.